

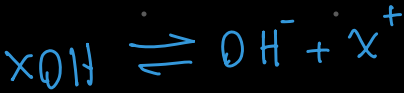
Equilíbrio Ionico

ácido



$$K_a = \frac{[H^+][A^-]}{[HA]}$$

base

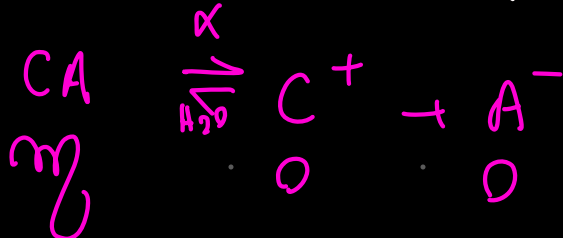


$$K_b = \frac{[OH^-][X^+]}{[XOH]}$$

generalização



$$\frac{[C^+][A^-]}{[CA]} = K_i$$

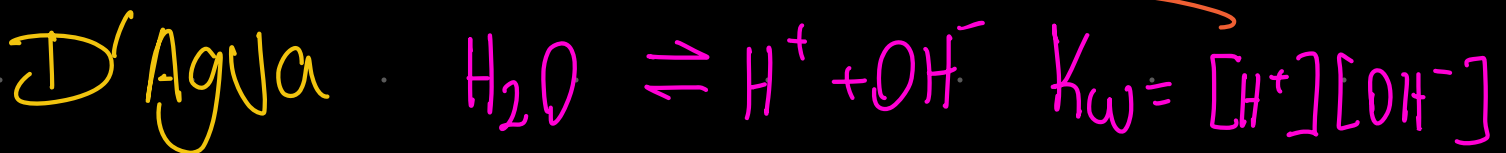


$$\begin{array}{ccc} -m\alpha & +m\alpha & +m\alpha \\ m(1-\alpha) & m\alpha & m\alpha \end{array}$$

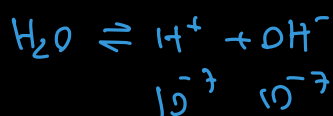
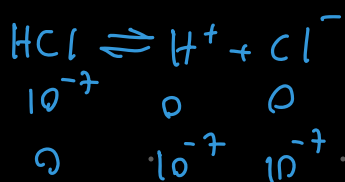
$$K_i = \frac{m^2 \alpha^2}{m(1-\alpha)} = \frac{m\alpha^2}{(1-\alpha)}$$

Para eletrólitos Fracos ($\alpha \ll 1$) $K_i = m\alpha^2$

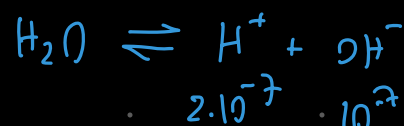
Lei de Ostwald reduzir $\downarrow m\alpha^2 \uparrow = K_i$
 reduzir m aumenta α em solução



HCl em água $[HCl] = 2 \cdot 10^{-7} \alpha = 200 \%$



Imediatamente pH é $-\log 2 \cdot 10^{-7}$



$$K_w = 10^{-14} = \frac{[H^+][OH^-]}{[H_2O]} = (2 \cdot 10^{-7} - x)(10^{-7} - x)$$

$$0 = x^2 - 3 \cdot 10^{-7}x + 10^{-14}$$

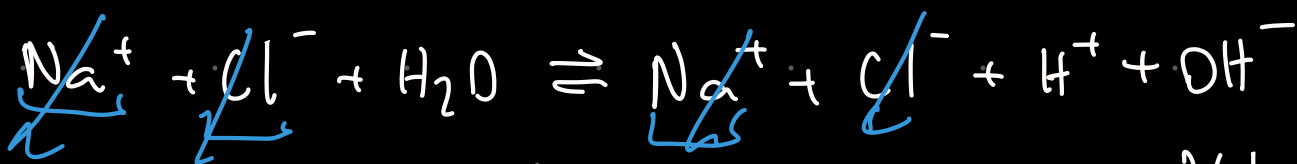
$$x = \left(\frac{3 \pm \sqrt{5}}{2} \right) \cdot 10^{-7}$$

$$x = \frac{3 - \sqrt{5}}{2} \cdot 10^{-7}$$

$$pH = -\log \left(2 \cdot 10^{-7} - \left(\frac{3 - \sqrt{5}}{2} \right) \cdot 10^{-7} \right)^2$$

Hidrólise de Sais

quando o sal é representado pelo produto de água.



não ocorre hidrólise do sal $K_h = K_w$

• Sal de Ácido Fraco e Base Fraca



$$K_h = \frac{[NH_4OH][H^+]}{[NH_4^+]} \cdot \frac{[OH^-]}{[OH^-]} = \frac{[NH_4OH][H^+][OH^-]}{[NH_4^+][OH^-]} = \frac{K_w}{K_b}$$

Somando Ion ∞ eletrolito fraco Ionico

Sal de ACID Fraco e Base Fica

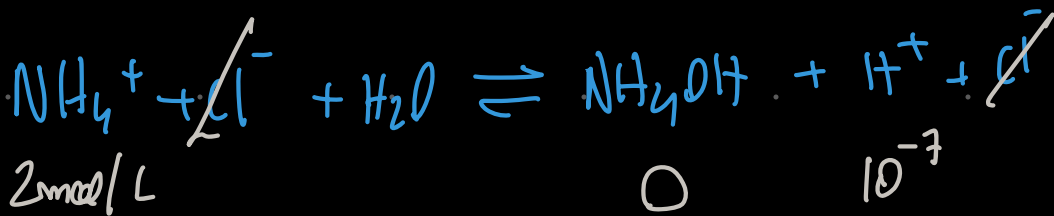


$$K_h = \frac{[\text{NH}_4\text{OH}][\text{HCN}]}{[\text{NH}_4^+][\text{CN}^-]} \cdot \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}^+][\text{OH}^-]} = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}^+][\text{OH}^-]} = 1$$

$$\frac{[\text{NH}_4\text{OH}]}{[\text{NH}_4^+][\text{OH}^-]} \cdot \frac{[\text{HCN}]}{[\text{H}^+][\text{CN}^-]} \cdot [\text{H}^+][\text{OH}^-] = \frac{K_w}{K_a \cdot K_b}$$

Em uma hidrólise de NH_4Cl qual o pH final

$$K_a = \text{muito Grande} \quad K_b = 2 \cdot 10^{-5} \quad \frac{K_w}{K_b} = K_h \Rightarrow \frac{10^{-14}}{2 \cdot 10^{-5}} \Rightarrow K_h = 5 \cdot 10^{-10}$$



2mol/L

0

10^{-7}

$$\frac{[\text{H}^+][\text{NH}_4\text{OH}]}{[\text{NH}_4^+]}$$

-x

+x

+x

$[\text{NH}_4^+]$

2-x

x

$10^{-7} + x$

$$5 \cdot 10^{-10} = \frac{x^2}{2-x}$$

K_h muito pouco
muito pouco

$$\frac{x^2}{2} = 5 \cdot 10^{-10}$$

$$\text{pH} = 5$$

$[H^+]$ e $[OH^-]$ tem que respeitar equilíbrio iônico
da água, vão reagir entre si até
subir, descer ou de um em particular